Deepak Roshan A

Cuddalore, Tamil Nadu | +91 9361940512 | deepakroshan380@gmail.com | linkedin.com/in/deepakroshan-adr | github.com/deepakroshan11 | deepakroshan.is-a.dev

Professional Summary

AI Engineer specializing in **Python**, **Machine Learning**, and **AWS**, with hands-on experience building and deploying **end-to-end AI systems**. Developed production-like applications involving **computer vision**, **NLP**, and real-time data pipelines. Skilled in model development, backend integration, and cloud deployment, delivering scalable solutions for real-world problems.

Education

Sri Krishna College of Technology **B.E. in Artificial Intelligence and Machine Learning** | 2022 – 2026 *CGPA: 7.0*

Technical Skills

Frontend: HTML5, CSS3, JavaScript, TypeScript, React,Tailwind CSS, Three.js
Programming: Python (NumPy, Pandas, SciPy)
AI/ML: TensorFlow, OpenCV, CNNs, Transformers
Backend: Node.js, FastAPI, Flask, REST APIs, Socket.IO
Cloud: AWS (EC2, Lambda, CloudWatch, SNS), Docker, Hugging Face Spaces
Databases: PostgreSQL (SQL, Supabase), Cloudinary
Systems & Tools: Linux, System Monitoring, GitHub Actions (CI/CD)

Certifications

AWS Educate – Cloud Computing Foundations | *Amazon Web Services* *2025*
Claude Code in Action | *Anthropic Academy* *2026*
Introduction to Computer Vision with OpenCV | *Coursera / OpenCV.org* *2024*

Work Experience

Cloud Operations Intern | *Appin Technology* *Apr 2025 – Jun 2025*

- Deployed applications on AWS (EC2, S3, Lambda) using Linux.
- Used shell scripts to automate basic deployment and system tasks.
- Monitored systems and logs using `cron`, `htop`.
- Optimized cloud resources for better performance.

Technical Projects

Findora – AI-Powered Lost and Found Platform | *React, FastAPI, MobileNetV3, MiniLM* **– GitHub | Live Demo** *2026*

- Full-stack platform (React, FastAPI, Supabase) with Google OAuth + email auth for lost/found item matching.
- Used MobileNetV3 and MiniLM for item matching across vision and text features.
- Designed a scoring system to rank matches and trigger alerts for high-confidence results.
- Deployed on Vercel (frontend) and Render (backend) with Cloudinary CDN for image optimization.

AI-Powered ISL Translation with Emotion Awareness | *TensorFlow, MediaPipe, Flask* **– GitHub | Live Demo** *2025*

- Built a real-time ISL translation system using MediaPipe and a TensorFlow CNN with ~95% accuracy.
- Implemented emotion detection using facial landmarks with temporal smoothing.
- Designed a cloud-compatible system for real-time processing and deployment.
- Developed a speech-to-sign avatar using Three.js and speech recognition.

CloudGuard AI – Intelligent Cloud Monitoring & Self-Healing | *AWS, Python, ML, Streamlit* **– GitHub | Live Demo** *2024*

- Monitored real-time EC2 metrics using CloudWatch.
- Used Isolation Forest to detect anomalies in system behavior.
- Automated self-healing using AWS Lambda and SNS alerts.
- Built a Streamlit dashboard for metrics, anomalies, and recovery.

VOXAI – Neural Voice Clone Studio | *Python, FastAPI, ChatterboxTTS* **– GitHub | Live Demo** *2025*

- Built a voice cloning system that generates speech from short audio samples.
- Used ChatterboxTTS for zero-shot voice synthesis with accent preservation.
- Designed an audio processing pipeline for quality enhancement.
- Deployed using FastAPI with a web interface on Hugging Face Spaces.